YouTube video 4: Balancing chemical equations, step by step (Class 10)

Format: whiteboard or screen-record · 8–10 min · Class 10 Ch 1 (CBSE + GSEB)

Title options

1. Balancing Chemical Equations in 5 Easy Steps | Class 10 Science Ch 1 | CBSE & GSEB
2. Never Get Balancing Wrong Again: The Atom-Count Table Method | Class 10
3. Class 10 Chemistry: Balance ANY Equation (Fe + H_2O, Al + O_2, Propane)

Thumbnail text: "BALANCE ANY EQUATION" + "5 STEPS" · show Fe + H_2O → Fe_3O_4 + H_2 with a red ✗ and the balanced version with a green ✓.

Script

[0:00 Hook] "If you've ever balanced an equation, got it 'almost right' and still lost the mark, this video is for you. I'll show you a 5-step method that works every single time. We'll do three equations together, and then I'll give you three to try."

[0:20 Why balance at all?] "Atoms can't appear or disappear in a reaction. That's the law of conservation of mass. So the number of each atom on the left **must** equal the number on the right. A balanced equation is just us counting honestly."

[0:50 The 5 steps] *(write on board)*

1. Write the skeletal equation.
2. Make an atom-count table: element, number on the left, number on the right.
3. Start with the **biggest formula** (the one with the most atoms).
4. Change **only the numbers in front** (coefficients). Never touch the small numbers (subscripts).
5. Re-count everything, then add states (s), (l), (g), (aq) if the question asks.

[1:40 Example 1: iron and steam] "Fe + H_2O → Fe_3O_4 + H_2. Let's count. Fe: 1 and 3. H: 2 and 2. O: 1 and 4. The biggest formula is Fe_3O_4. It has 4 oxygens, so put 4 in front of H_2O: Fe + $4H_2O$ → Fe_3O_4 + H_2. Now hydrogen: 8 on the left, 2 on the right. Put 4 in front of H_2: Fe + $4H_2O$ → Fe_3O_4 + $4H_2$. Finally iron: 1 on the left, 3 on the right, so 3Fe. **$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.** Re-count: Fe 3 = 3, H 8 = 8, O 4 = 4 ✓. With states: $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$."

[3:30 Example 2: the LCM trick] "Al + O_2 → Al_2O_3. Oxygen is 2 on the left and 3 on the right. When you see 2 and 3, think LCM, which is 6. $3O_2$ gives 6, and $2Al_2O_3$ gives 6. Now aluminium on the right is 2 × 2 = 4, so 4Al. **$4Al + 3O_2 \rightarrow 2Al_2O_3$** ✓"

[5:00 Example 3: burning a fuel (C, H, O)] "Propane, the gas in some cylinders: C_3H_8 + O_2 → CO_2 + H_2O. Rule for fuels: **C first, then H, oxygen last.** C: 3, so $3CO_2$. H: 8, so $4H_2O$. Now count oxygen on the right: 3 × 2 + 4 × 1 = 10, so $5O_2$. **$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$** ✓"

[6:30 Three mistakes that cost marks]

1. Changing H_2O to H_2O_2 to "make it balance". That's a different substance!
2. Forgetting to re-count after the last change.
3. Skipping states when the question says "with physical states".

[7:30 Your turn] "Pause the video and try these three. The answers are in the pinned comment. (1) $Na + H_2O \rightarrow NaOH + H_2$ (2) $KClO_3 \rightarrow KCl + O_2$ (3) $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$"

[8:00 Close] "If this helped, share it with one friend who's scared of chemistry 🙂 Next video: how to write a 3-mark answer that actually gets 3 marks. See you there!"

Description

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Balancing chemical equations made simple, for Class 10 Science Chapter
1 (CBSE & GSEB).
Learn the 5-step atom-count method, the LCM trick, and the C-H-O rule
for fuels, with 3 solved examples and 3 practice questions.

⏱️ Chapters
0:00 Why balancing trips everyone up
0:50 The 5-step method
1:40 Example 1: Fe + H₂O
3:30 Example 2: Al + O₂ (LCM trick)
5:00 Example 3: Propane (C, H, then O)
6:30 3 mistakes that cost marks
7:30 Practice questions

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Tags: balancing chemical equations, class 10 science chapter 1, chemical reactions and equations, cbse class 10 science, gseb std 10 science, how to balance equations, class 10 chemistry, board exam 2027

Pinned comment

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Answers 👇
(1) 2Na + 2H₂O → 2NaOH + H₂
(2) 2KClO₃ → 2KCl + 3O₂
(3) 2C₂H₆ + 7O₂ → 4CO₂ + 6H₂O
How many did you get right? Tell me in the comments 🙂
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Shorts cuts (vertical, under 60 s)

1. **"The LCM trick for balancing"**: Example 2 only (0:00–0:45).
2. **"Never do THIS while balancing"**: the $H_2O \rightarrow H_2O_2$ mistake, plus the correct way (30 s).